Problem 4

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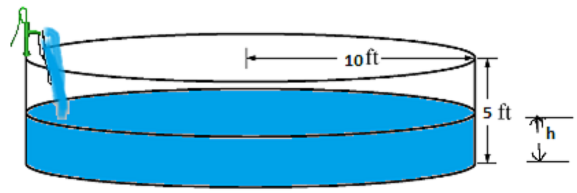
Problem 1 *A cylindrical swimming pool with a radius of 10 feet and height of 5 feet is being filled with water. How fast is the water being pumped into the pool (in* ft^3/min*) if the water level is increasing at a rate of 6 in/minute? Make sure to draw a picture to represent the situation.*



Explanation.

$$
\begin{aligned}
\frac{dh}{dt} &= 6\text{in/min} = .5\text{feet/min} \\
V &= \pi r^2 h = \pi(100)h \\
\frac{dV}{dt} &= 100\pi \frac{dh}{dt} \\
\frac{dV}{dt} &= 100\pi(.5) = 50\pi \text{ft}^3/\text{min}
\end{aligned}
$$